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Synergistic chemo-photothermal therapy using doxorubicin-loaded gold nanorods for enhanced apoptosis in lung cancer cells

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Lung cancer remains one of the most lethal malignancies worldwide, largely due to late-stage detection, rapid progression, and resistance to conventional therapies. Although current modalities—surgery, chemotherapy, radiotherapy, targeted therapy, and immunotherapy—have improved outcomes, long-term survival rates remain unsatisfactory. Photothermal therapy (PTT), which employs near-infrared (NIR) light to induce localized hyperthermia, has recently emerged as a promising complement to standard treatments by enhancing drug delivery and overcoming resistance. In this work, we investigated a combinatorial strategy using doxorubicin (DOX)-loaded gold nanorods functionalized with thiolated β -cyclodextrin (AuNRs@S- β -CD-DOX) in conjunction with 808 nm NIR laser irradiation in A549 human lung cancer cells. Apoptosis was evaluated through gene expression profiling, TUNEL assay, and Western blot analysis. Mechanistically, this triple approach activated both intrinsic (caspase-9) and extrinsic (caspase-8) apoptotic pathways, as evidenced by cleavage of procaspases and robust caspase-3 activation. Strikingly, the combined treatment (DOX 0.078 μ M + AuNRs@S- β -CD + laser) induced apoptosis in nearly 60% of cells, far surpassing single or dual treatments, while operating at a DOX dose substantially below its IC₅₀ value. The synergistic effect was attributed to enhanced intracellular DOX release mediated by AuNR photothermal heating and increased membrane permeability upon laser exposure. These findings demonstrate that AuNR-assisted chemo-photothermal therapy can overcome drug resistance and amplify apoptosis, providing a rational, targeted, and effective strategy for advancing lung cancer treatment.

Keywords Photothermal therapy, Doxorubicin, Gold nanorods, Lung cancer, Drug resistance, Apoptosis, Synergistic therapy

Lung cancer, comprising non-small cell lung cancer (NSCLC, ~ 85%) and small cell lung cancer (SCLC, ~ 15%)^{1,2}, is the most commonly diagnosed malignancy and a leading cause of cancer-related mortality worldwide^{1,3–6}. Despite advances in surgery, chemotherapy, radiotherapy, targeted therapy, and immunotherapy, the 5-year survival rate remains below 20%, largely due to late-stage diagnosis, metastatic spread, and the development of resistance to conventional treatments⁵. These challenges highlight the urgent need for more effective and selective therapeutic approaches, and in recent years, nanotechnology-based strategies such as photothermal therapy (PTT) have attracted growing attention^{7–10}.

Current treatment modalities for lung cancer encompass surgery, chemotherapy, radiotherapy, immunotherapy, and emerging PTT methods, each with unique advantages and limitations^{11,12}. Surgical resection is often the preferred treatment for early-stage NSCLC; however, due to late presentation, only 20–25% of patients are eligible¹². In contrast, surgery has little value in SCLC due to its aggressive and systemic nature. Beyond restricted applicability, surgery carries inherent morbidity, limited benefit for metastatic or inoperable tumors, and risk of recurrence^{2,13}.

Chemotherapy remains the backbone of lung cancer treatment, with agents such as cisplatin, carboplatin, and doxorubicin (DOX) frequently employed^{14,15}. While these agents induce apoptosis by damaging DNA in rapidly

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dividing cells, they are severely limited by non-specific toxicity, narrow therapeutic index, and most importantly, the emergence of drug resistance^{16–20}. DOX resistance, in particular, is a major obstacle in clinical oncology. Mechanistically, this resistance arises through multiple interconnected processes, including overexpression of ATP-binding cassette (ABC) efflux transporters such as P-glycoprotein (MDR1), impaired intracellular accumulation of DOX, enhanced DNA repair mechanisms, activation of pro-survival signaling pathways, and inhibition of apoptosis through caspase suppression^{19,20}. Additionally, redox imbalances and elevated NRF2-mediated antioxidant responses contribute to the refractory phenotype of many lung cancer cells. As a result, achieving clinically effective outcomes often necessitates high DOX dosages, which in turn exacerbate systemic cardiotoxicity and other dose-limiting side effects.

Radiotherapy is frequently combined with chemotherapy in locally advanced NSCLC and SCLC to maximize cytotoxic efficacy¹¹. While it induces DNA double-strand breaks and enhances tumor control, its non-selectivity also causes collateral damage to healthy tissues and may lead to pneumonitis or fibrosis^{16,17}.

Photothermal therapy (PTT) represents a minimally invasive alternative that harnesses light-absorbing agents to generate localized hyperthermia under near-infrared (NIR) irradiation, thereby selectively ablating tumor tissue. Plasmonic nanoparticles, such as gold nanorods (AuNRs), are particularly attractive as photothermal agents due to their strong absorption at 808 nm, efficient photothermal conversion, and tunable optical properties. Irradiation of AuNRs at the appropriate wavelength results in rapid heating of the nanoparticles with minimal impact on the surrounding aqueous environment, enabling spatially confined cytotoxicity. The highly localized nature of PTT offers the additional advantage of precise tumor targeting^{7–10,21–27}.

Nevertheless, PTT as a monotherapy faces critical limitations. Light penetration is restricted by tissue absorption and scattering, reducing its effectiveness in deep-seated tumors, and incomplete thermal ablation often allows for tumor regrowth^{28,29}. To address these shortcomings, researchers have increasingly explored combinatorial strategies that integrate PTT with chemotherapy, capitalizing on the synergistic effect of hyperthermia and cytotoxic drugs. Hyperthermia has been shown to enhance cell membrane permeability, increase intracellular drug uptake, and sensitize resistant cancer cells by interfering with heat-shock and DNA repair pathways^{30,31}. This combinatorial effect is particularly relevant in overcoming DOX resistance, as mild hyperthermia can facilitate endosomal escape, reduce drug efflux activity, and re-engage apoptotic pathways otherwise suppressed in resistant phenotypes^{19,20,32}.

In this context, nanoscale delivery systems offer further therapeutic advantages. Their small size enables efficient endocytosis and intracellular trafficking, while surface functionalization enhances stability, drug-loading capacity, and biocompatibility^{5,33}. Gold nanoparticles, in particular, have emerged as versatile therapeutic platforms due to their unique plasmonic behavior, tunable size and shape, high biocompatibility, and ease of functionalization^{34–25}. Among these, AuNRs stand out as promising candidates for combined chemo-photothermal applications. However, their aggregation in biological environments and subsequent loss of optical properties present challenges that must be mitigated through surface modification with stabilizing agents^{35–42}.

Doxorubicin remains a cornerstone chemotherapeutic agent for both hematological and solid tumors^{14,43}. While its potent DNA intercalation and topoisomerase II inhibition confer broad-spectrum anticancer activity, its clinical utility in lung cancer is hindered by the mechanisms of resistance outlined above¹⁴. Therefore, a rational strategy is to integrate DOX delivery with AuNR-mediated PTT, thereby achieving synergistic activation of apoptotic pathways while minimizing systemic toxicity. Importantly, this dual-modality approach has the potential not only to improve therapeutic outcomes but also to mechanistically overcome DOX resistance by restoring apoptosis signaling through both the intrinsic (mitochondrial, caspase-9) and extrinsic (death receptor, caspase-8) pathways.

Given these considerations, the present study was designed to investigate the effects of DOX, AuNRs, and NIR laser irradiation individually and in combination in A549 human lung cancer cells. To elucidate the molecular basis of treatment efficacy, we assessed apoptosis by profiling key gene expression changes, performing TUNEL assays, and validating caspase activation at the protein level via Western blotting. This comprehensive genetic and mechanistic analysis aims to advance our understanding of how chemo-photothermal therapy can effectively overcome DOX resistance and promote apoptosis in drug-refractory lung cancer cells.

Despite these efforts, DOX resistance remains widespread across multiple cancer cell types, including lung carcinoma. A key mechanism involves membrane-bound transporters that actively expel DOX, thereby preventing nuclear accumulation and inhibition of topoisomerase II. This phenotype has been specifically reported in the A549 cell line, where a Keap1 gene mutation (G333C) leads to NRF2 overactivation and subsequent upregulation of multidrug resistance proteins such as P-glycoprotein (P-gp) and lung resistance-related protein (LRP), both of which reduce intracellular DOX concentration⁴⁴. Furthermore, anti-apoptotic factors including Bcl-2 and FLIP contribute to suppression of caspase-mediated cell death, reinforcing the resistant phenotype¹⁵. Understanding these molecular determinants of resistance underscores the importance of strategies, such as chemo-photothermal therapy, that can circumvent efflux activity and restore apoptotic signaling.

Results

In this study, we employed an 808 nm near-infrared (NIR) laser with a power density of 3.6 W/cm². When AuNRs@S-β-CD was introduced into the culture medium, laser irradiation required approximately 40 s to raise the temperature of the medium to 47 °C. In contrast, under DOX (0.078 μM) + AuNRs@S-β-CD conditions, the same temperature threshold was reached after about 50 s of irradiation. Interestingly, following the initial irradiation of the DOX (0.078 μM) + AuNRs@S-β-CD-treated medium, the rate of temperature elevation nearly doubled during subsequent irradiations, suggesting enhanced thermal responsiveness of the system. Figure 1 summarizes these thermal dynamics under the described experimental conditions.

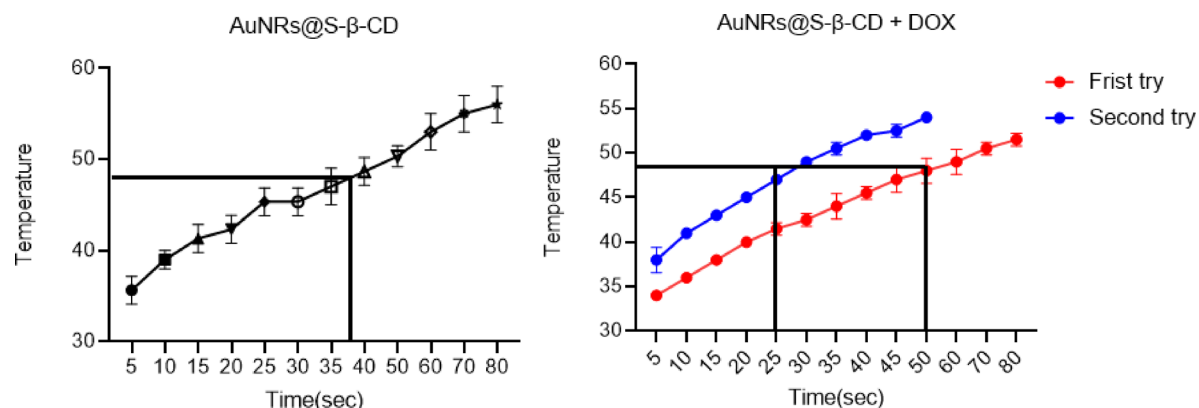


Fig. 1. Time-dependent temperature profile using AuNRs@S-β-CD and in combination with DOX.

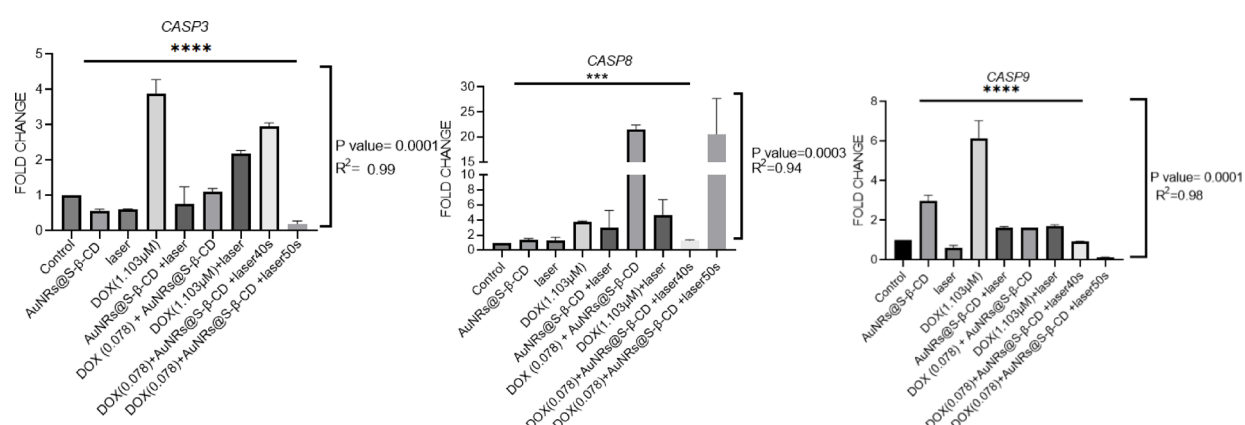


Fig. 2. Expression profiles of key genes in the cell apoptosis show significant variations in different groups. All experiments were performed with at least four independent biological replicates, each measured in at least six technical replicates. Data are presented as mean $\pm$ standard deviation (SD).

Subsequently, we assessed the expression profiles of key apoptotic markers—including caspase-3, caspase-8, and caspase-9—across different combinatorial treatment groups to elucidate the molecular mechanisms underlying the observed cytotoxic effects (Fig. 2).

The expression of the CASP3 gene under AuNRs@S-β-CD, laser irradiation, and PTT (AuNRs@S-β-CD + laser) conditions was lower than that observed in the control group. By contrast, treatment with high-dose DOX (1.307 μ M) led to a marked increase in CASP3 expression relative to the control. The combination of low-dose DOX (0.078 μ M) with AuNRs@S-β-CD resulted in CASP3 expression levels comparable to those of the control group. Notably, DOX (1.307 μ M) combined with laser irradiation induced an increase in CASP3 expression greater than that of the control, but still lower than DOX alone. During chemo-photothermal treatment (DOX 0.078 μ M + AuNRs@S-β-CD + laser), CASP3 expression rose above control levels within the first 40 s of irradiation, when DOX release had begun but was incomplete. However, after 50 s of irradiation—corresponding to full drug release—a reduction in expression below control levels was observed. All observed changes were statistically significant ($p < 0.05$).

For CASP8, gene expression did not differ significantly between AuNRs@S-β-CD or laser-only treatments and the control. Interestingly, DOX (1.307 μ M) combined with laser irradiation decreased CASP8 expression, whereas treatment with DOX alone, DOX (0.078 μ M) + AuNRs@S-β-CD, or AuNRs@S-β-CD + laser each resulted in increased expression relative to the control. Under chemo-photothermal conditions, CASP8 expression remained unchanged during the first 40 s of irradiation, when DOX release was still incomplete. However, after 50 s of laser exposure, coinciding with complete DOX release, CASP8 expression rose sharply, showing a highly significant increase compared with the control ($p < 0.05$).

CASP9 expression analysis revealed that DOX (1.307 μ M) alone produced the greatest increase compared with the control. AuNRs@S-β-CD alone also enhanced CASP9 expression, while the binary combinations DOX (1.307 μ M) + laser, DOX (0.078 μ M) + AuNRs@S-β-CD, and AuNRs@S-β-CD + laser each yielded similar levels of increased expression relative to the control. In contrast, laser irradiation alone reduced CASP9 expression below control levels. Under chemo-photothermal treatment, CASP9 expression showed no significant change

during the first 40 s of irradiation. Yet, after 50 s of laser exposure and full DOX release, a significant decrease in CASP9 expression was observed ($p < 0.05$).

TUNNEL assay

The TUNEL assay results presented in Fig. 3 indicate that apoptosis levels in cells treated with AuNRs@S-β-CD alone were comparable to those of the control group, confirming that the nanocarrier itself does not induce apoptosis. Laser irradiation and PTT (AuNRs@S-β-CD + laser) produced nearly identical effects, both eliciting only a modest increase in apoptosis relative to the control. Treatment with high-dose DOX (1.307 μM) induced less than 40% apoptosis, corresponding to an approximate 20% increase compared with control cells. When DOX (1.307 μM) was combined with laser irradiation, apoptosis levels rose further, representing a ~ 30% increase compared with the control group.

In contrast, treatment with low-dose DOX (0.078 μM) + AuNRs@S-β-CD at the acidic pH characteristic of the tumor microenvironment facilitated drug release and significantly elevated apoptosis relative to the control condition. Notably, chemo-photothermal therapy (DOX 0.078 μM + AuNRs@S-β-CD + laser) produced the most pronounced apoptotic effect: after 50 s of irradiation, complete release of DOX from the nanoparticles was achieved, resulting in a 60% apoptosis rate. This synergistic outcome highlights the remarkable therapeutic potential of the platform, achieving high apoptotic efficacy at a DOX dose far below its IC50 threshold. The quantitative comparisons among different treatment groups are summarized in the graphical representation, which underscores the superior effectiveness of the combined approach.

Western blot results

We examined three major caspase proteins, each of which exists in two forms: the inactive precursor (procaspase) and the cleaved, active form. The results are summarized in Fig. 4.

Caspase-8

As shown in Fig. 4 (A, C), the 57 kDa bands correspond to the uncleaved procaspase-8. Activation occurs through proteolytic cleavage, generating lower-molecular-weight fragments (43 kDa and 18 kDa) that represent the functional caspase-8 isoforms. The most pronounced activation was observed in cells treated with DOX (0.078 μM) + AuNRs@S-β-CD and in the full chemo-photothermal regimen (DOX 0.078 μM + AuNRs@S-β-CD + laser). Under these conditions, nearly all procaspase-8 was converted into the active 43 kDa and 18 kDa fragments, with levels markedly elevated compared with the control. In contrast, control cells displayed only the 43 kDa active fragment, with no evidence of the 18 kDa band, while a substantial portion of the protein remained in the inactive 57 kDa state. AuNRs@S-β-CD alone did not produce detectable activation and, in fact, yielded reduced protein levels compared to the control. Laser treatment alone induced partial activation,

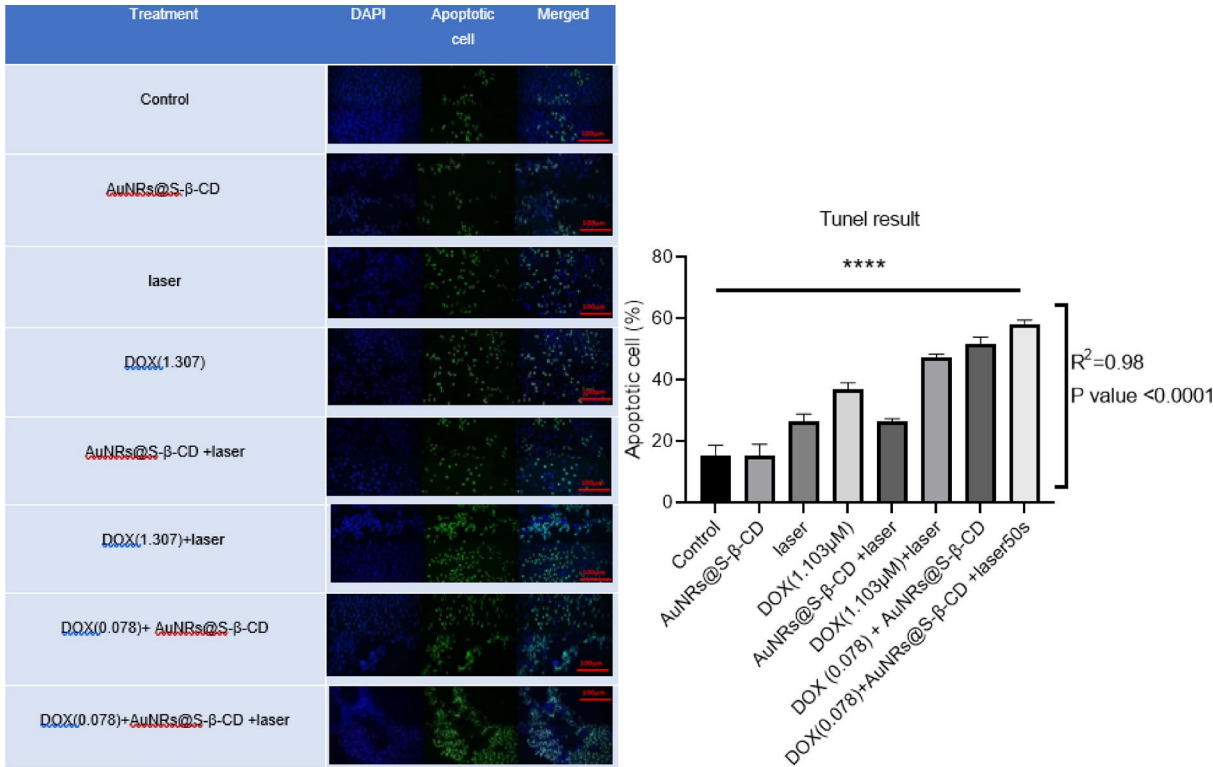


Fig. 3. The cell apoptosis was monitored in intended groups using TUNNEL assay. For the TUNEL assay, each sample was plated in duplicate, and representative images are shown.

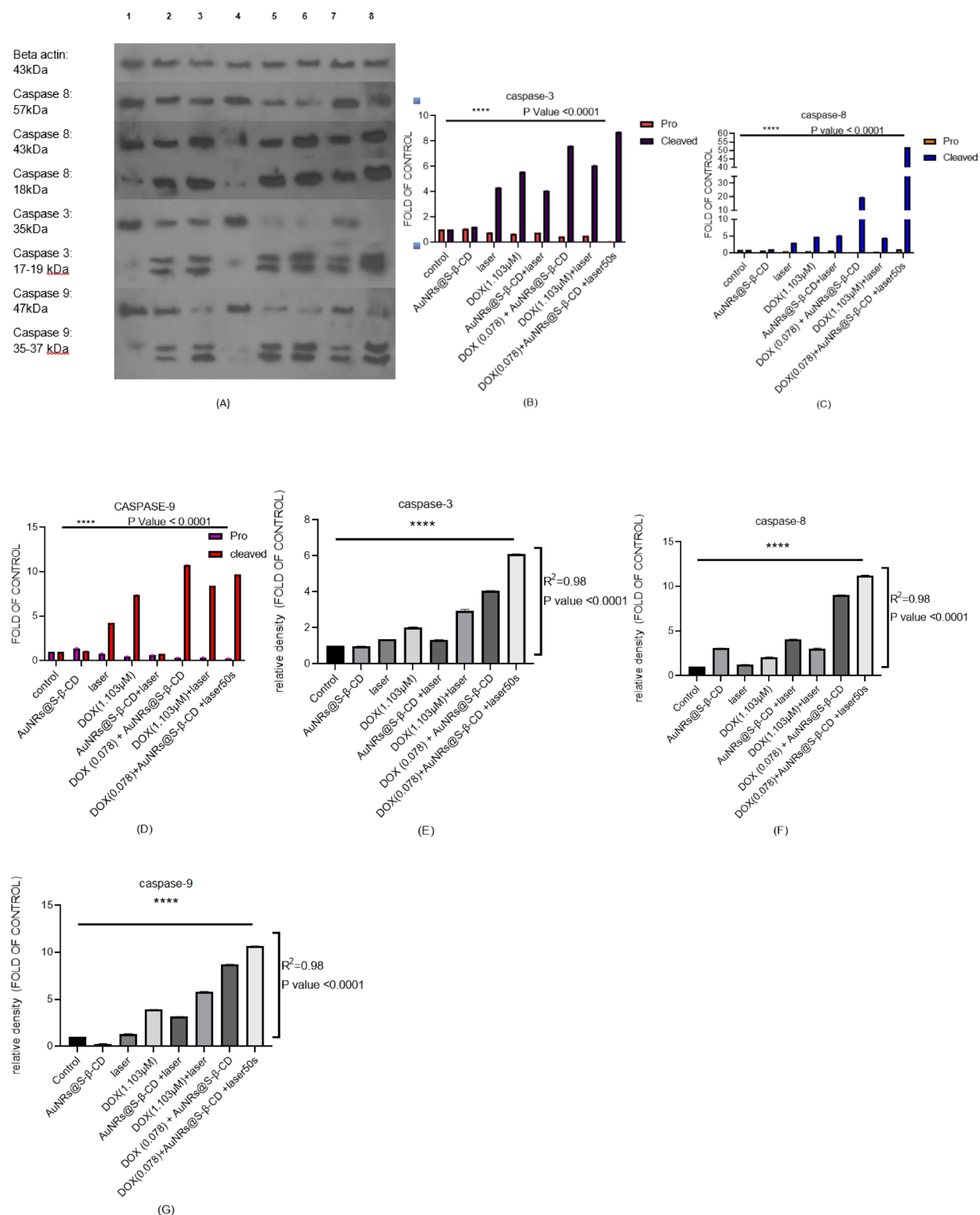


Fig. 4. (A) The picture of gel: (1) Control, (2) laser, (3) DOX (1.307), (4) AuNRs@S-β-CD, (5) DOX (1.307) + laser, (6) DOX (0.78) + AuNRs@S-β-CD, (7) AuNRs@S-β-CD + laser, (8) DOX (0.78) + AuNRs@S-β-CD + laser. (B), (C), and (D): The comparative graph between relative density of cleaved and uncleaved Caspase proteins. (E), (F), and (G): The comparative graph between total densities of Caspases by different treatments. Western blot analyses were performed multiple times to ensure reproducibility, while all other procedures were conducted once for each sample.

with both active fragments present but accompanied by residual procaspase. DOX (1.307 μM) induced both active fragments but retained inactive protein, while DOX (1.307 μM) + laser closely resembled this profile. PTT (AuNRs@S- β -CD + laser) enhanced caspase-8 activation but to a lesser extent than the low-dose DOX-based regimens.

Caspase-3

As illustrated in Fig. 4 (A, B), the 35 kDa band represents the uncleaved procaspase-3, whereas the 19 kDa and 17 kDa fragments correspond to its active forms. The greatest caspase-3 activation occurred in the DOX (0.078 μM) + AuNRs@S- β -CD group and under chemo-photothermal treatment (DOX 0.078 μM + AuNRs@S- β -CD + laser), where nearly all procaspase was converted into the active fragments, showing significantly higher levels than the control. In the control group, a portion of the protein remained in the inactive state, and no active fragments were detected in AuNRs@S- β -CD-only treatment. Laser irradiation alone produced both active fragments, although inactive protein was still present. High-dose DOX (1.307 μM) induced both active forms but retained a significant amount of inactive procaspase. Combination with laser (DOX 1.307 μM + laser) resulted in nearly complete activation, while PTT (AuNRs@S- β -CD + laser) produced a caspase-3 profile very similar to that of DOX (1.307 μM).

Caspase-9

As shown in Fig. 4 (A, D), the 47 kDa band represents procaspase-9, while the 37 kDa and 35 kDa bands represent the cleaved, active forms. The strongest caspase-9 activation was observed in DOX (0.078 μM) + AuNRs@S- β -CD and in the chemo-photothermal treatment group, where nearly all procaspase was converted to its active fragments. A small residual amount of the inactive 47 kDa band was still present. In the control group, inactive caspase-9 was partially retained, whereas AuNRs@S- β -CD alone produced no active fragments and even reduced overall protein levels compared with the control. Laser irradiation alone, however, produced both active caspase-9 fragments, confirming partial activation.

Taken together, these Western blot findings demonstrate that the triple chemo-photothermal treatment robustly activates both initiator (caspase-8 and -9) and executioner (caspase-3) caspases, underscoring its capacity to simultaneously engage intrinsic and extrinsic apoptotic pathways.

Discussion

Despite initial responsiveness to chemotherapy, a large proportion of lung cancer patients ultimately experience tumor relapse, largely due to acquired resistance—particularly against agents such as doxorubicin (DOX)¹⁹. Mechanistically, this resistance often arises from enhanced drug efflux, impaired nuclear accumulation, and suppression of apoptotic signaling pathways. These limitations highlight the urgent need for innovative therapeutic strategies that can improve drug delivery, maximize efficacy, and minimize systemic toxicity²⁰.

Our findings demonstrate that a chemo-photothermal strategy combining low-dose DOX (0.078 μM), thiolated β -cyclodextrin-functionalized gold nanorods (AuNRs@S- β -CD), and 808 nm laser irradiation significantly augments apoptosis in A549 lung cancer cells. TUNEL assay results confirmed apoptosis rates of up to 60%, far exceeding the ~15% baseline in untreated controls and outperforming high-dose DOX (1.307 μM), which induced only ~37% apoptosis. Achieving such efficacy at a sub-IC₅₀ concentration underscores the therapeutic promise of nanoparticle-assisted drug delivery coupled with localized hyperthermia.

Mechanistically, the enhanced effect is mediated by synergistic activation of both extrinsic (caspase-8) and intrinsic (caspase-9) apoptotic pathways⁴⁴. Gene expression and Western blot analyses revealed robust cleavage of procaspase-8 into its active 43 kDa and 18 kDa fragments, accompanied by strong activation of caspase-3 and caspase-9. These findings suggest that laser-triggered heating accelerates DOX release from AuNRs@S- β -CD, leading to higher intracellular drug accumulation, circumvention of P-glycoprotein-mediated efflux, and restoration of apoptosis signaling³².

While AuNRs@S- β -CD alone showed no pro-apoptotic effect and laser irradiation alone induced only modest activity, the combination of DOX with laser irradiation significantly increased apoptosis (~47%), likely due to enhanced membrane permeability and reactive oxygen species (ROS) generation^{45–47}. The most striking outcome was observed under the triple combination, which produced nearly a 20-fold increase in caspase-8 expression, indicating potent activation of the death receptor-mediated pathway.

These results are consistent with previous reports that photothermal stimulation facilitates endosomal escape and intracellular drug release. Importantly, our study shows that chemo-photothermal therapy not only improves DOX performance but also reprograms apoptotic signaling, effectively overcoming resistance in A549 cells known to harbor KEAP1 mutations and elevated NRF2 activity, both of which promote drug efflux and survival⁴⁸.

Despite the promising outcomes, certain limitations must be acknowledged. The present study is restricted to in vitro experiments, and in vivo validation is necessary to evaluate pharmacokinetics, biodistribution, and potential off-target effects. Moreover, the limited penetration depth of NIR light (~1–2 cm) remains a challenge for treating deep-seated lung tumors. Future advances may rely on integrating this strategy with endoscopic or fiber-optic laser delivery systems to improve clinical applicability.

In summary, our findings highlight gold nanorod-assisted chemo-photothermal therapy as an effective and targeted approach to enhance apoptosis in drug-resistant lung cancer cells. By enabling reduced chemotherapy dosage, localized cytotoxicity, and dual-pathway apoptotic activation, this platform shows strong potential for translation into preclinical and ultimately clinical development.

Conclusion

In conclusion, this study demonstrates that the integration of doxorubicin-loaded, thiolated β -cyclodextrin-functionalized gold nanorods with near-infrared laser irradiation represents a highly promising strategy for combating lung cancer, particularly in resistant phenotypes. By enabling a synergistic interaction between localized hyperthermia and controlled drug release, this triple-modality approach not only enhanced the apoptotic response but also allowed effective treatment at doxorubicin concentrations far below the IC50 threshold, thereby reducing the likelihood of systemic toxicity. Mechanistic analyses confirmed that the therapy simultaneously activates both intrinsic (caspase-9-mediated) and extrinsic (caspase-8-mediated) apoptotic cascades, culminating in robust caspase-3 activation and extensive programmed cell death. Such dual-pathway activation is particularly relevant for overcoming drug resistance, where apoptotic signaling is often suppressed.

Importantly, the study underscores that the observed therapeutic gains are not attributable to individual components alone—neither AuNRs nor laser irradiation alone were sufficient to significantly enhance apoptosis. Rather, the combination of photothermal heating with nanoparticle-facilitated drug delivery creates a dynamic microenvironment that increases membrane permeability, promotes intracellular accumulation of doxorubicin, and circumvents P-glycoprotein-mediated drug efflux. This represents a mechanistically rational approach to reversing multidrug resistance in A549 lung cancer cells, which are characterized by NRF2 overactivation and impaired apoptotic responses.

From a translational perspective, these findings highlight several key opportunities and challenges. On one hand, gold nanorod-based chemo-photothermal therapy offers clear advantages in terms of precision, dose sparing, and molecular reprogramming of resistant cancer cells. On the other hand, clinical adoption will require addressing issues such as limited NIR penetration depth, potential nanoparticle accumulation in non-target tissues, and the need for scalable, reproducible nanomaterial synthesis. Future work should focus on validating these results in vivo, optimizing nanoparticle pharmacokinetics and biodistribution, and exploring integration with multimodal therapeutic platforms such as immunotherapy or image-guided delivery systems.

Taken together, this study provides compelling evidence that chemo-photothermal therapy, when implemented with a rationally engineered nanocarrier system, holds significant potential as a next-generation therapeutic strategy for lung cancer. By combining spatially localized photothermal ablation with chemotherapy at sub-cytotoxic doses, this platform paves the way toward safer, more effective, and resistance-overcoming interventions that could ultimately improve clinical outcomes in patients with refractory lung cancer.

Materials and methods

Materials

Cetyltrimethylammonium bromide (CTAB), sodium borohydride (NaBH_4), L-ascorbic acid, Silver Nitrate (AgNO_3), Gold (III) chloride solution ($\text{HAuCl}_4 \cdot 3\text{H}_2\text{O}$), β -cyclodextrin (β -CD), p-toluenesulfonyl chloride (TsCl), and thiourea were purchased from Sigma-Aldrich. Deionized water, Chloroform, Ethanol, Trizol reagent, DEPC-treated water, DMEM cell culture medium, Fetal bovine serum (FBS), Penicillin-streptomycin, Doxorubicin (Cell culture Solarbio), cDNA synthesis kit (Yekta Tajhiz), SYBR green, DMSO (for cell culture), Trypsin-EDTA (0.25%), Phosphate-buffered saline (PBS), Tris buffer, TUNEL Assay kit (Roche), DAPI (9542D-Sigma), Sodium deoxycholate, NaCl, EDTA, HCl, Phosphoric acid, Coomassie Blue G250, NP-40, Triton, APS (Ammonium Persulfate), Bromophenol blue, β -mercaptoethanol, Distilled water, Glycerol, PVDF membrane, Bis-acrylamide, Acrylamide, Tetramethylethylenediamine (TEMED), Caspase-8 (1C12) antibody, Caspase-3 (D3R6Y) Rabbit mAb, Actin (2A3): sc-517,582, Donkey anti-goat IgG-HRP, Caspase-9 antibody (human-specific), β -Actin (2A3): sc-517,582, Caspase-3 (D3R6Y) Rabbit mAb, Caspase-8 (1C12) Mouse mAb, ECL advanced reagents, Tween 20 (0.1% v/v). Human lung cancer cell line A549 was obtained from the Pasteur Institute of Iran.

Methods

Synthesis of AuNRs@S- β -CD-DOX nanocomposite

Gold nanorods (AuNRs) were synthesized via a seed-mediated growth method. Briefly, gold seed particles were prepared by reducing HAuCl_4 with NaBH_4 in the presence of CTAB, followed by the addition of the seed solution to a growth solution containing HAuCl_4 , CTAB, AgNO_3 , and ascorbic acid. The resulting AuNRs were purified by centrifugation. Mono-6-thio- β -cyclodextrin (β -CD-SH) was synthesized in a multi-step process: β -cyclodextrin was first functionalized with p-toluenesulfonyl chloride (β -CD-OTs), followed by nucleophilic substitution with thiourea and subsequent hydrolysis to yield β -CD-SH. The AuNRs were then functionalized with β -CD-SH by incubating the AuNRs with the thiolated β -cyclodextrin overnight. The resulting AuNRs@S- β -CD nanocomposite was purified via centrifugation to remove excess β -CD-SH. This functionalization step provides a platform for further drug loading, such as doxorubicin (DOX), leveraging the host-guest interaction capability of β -cyclodextrin for targeted drug delivery applications⁴⁹.

The successful synthesis of AuNRs, AuNRs@S- β -CD, and AuNRs@S- β -CD-DOX was verified by UV-vis spectroscopy. The UV-vis absorption spectra of AuNRs, AuNRs@S- β -CD, and AuNRs@S- β -CD-DOX are shown in Fig. 5. The AuNRs exhibited characteristic transverse and longitudinal plasmon bands at 514 nm and 726 nm, respectively. Coating with thiolated β -cyclodextrin (HS- β -CD) caused a significant red-shift of the longitudinal band to 832 nm, confirming successful conjugation. Subsequent drug loading with doxorubicin (DOX) resulted in a further red-shift to 845 nm, indicating the successful attachment of the drug.

TEM and FESEM analysis confirmed the successful formation of the nanostructures. The TEM images, shown in Fig. 6, showed that the AuNR core structure was preserved after coating with HS- β -CD. Furthermore, TEM images of the final AuNRs@S- β -CD-DOX (Fig. 7) nanocomposite verified the successful loading of the

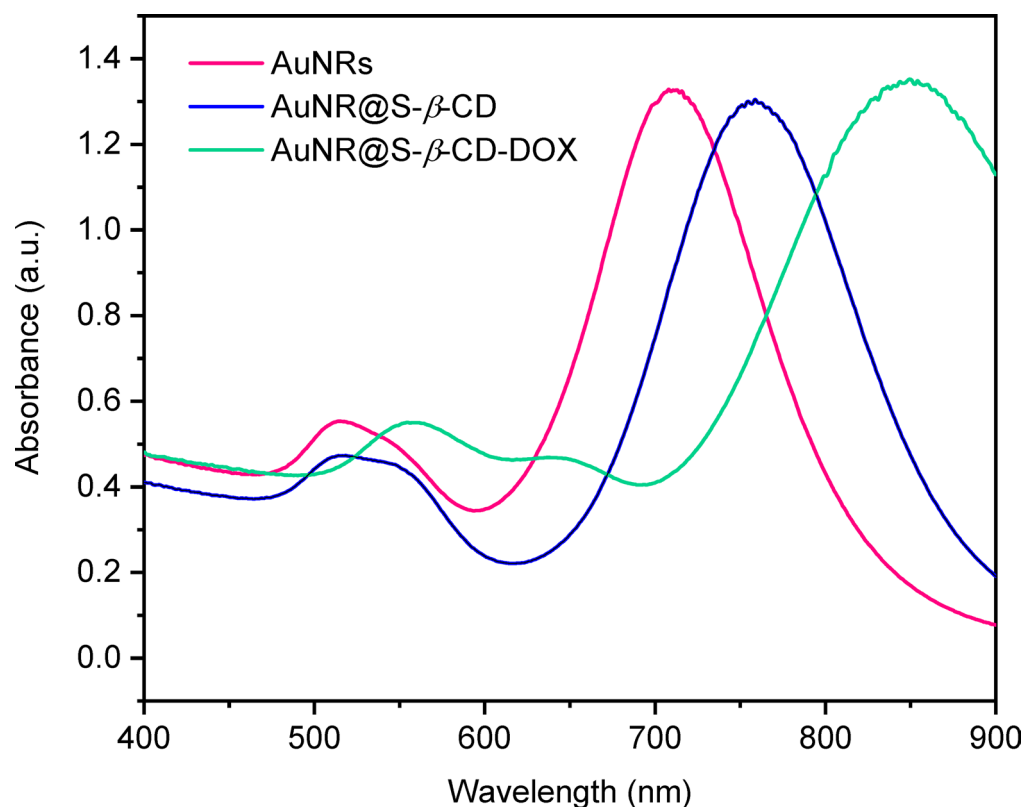


Fig. 5. The UV – vis absorption spectra of AuNRs, AuNRs@S- β -CD, and AuNRs@S- β -CD-DOX.

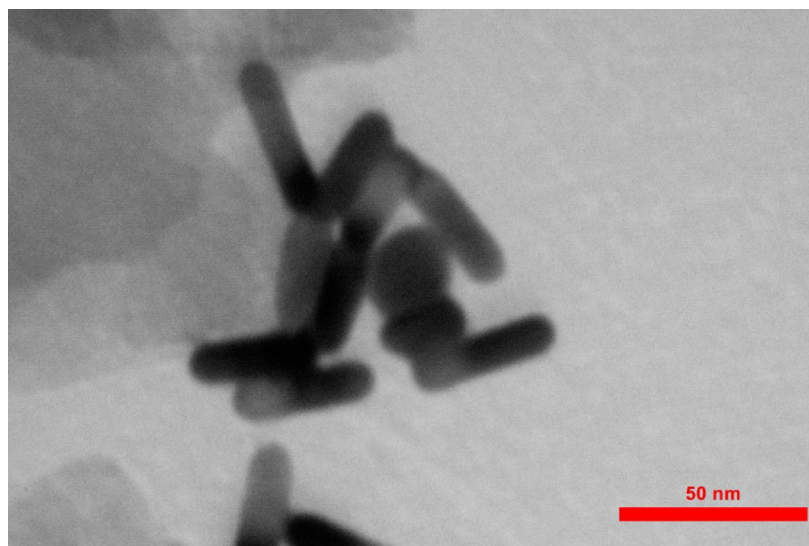


Fig. 6. TEM image of AuNR@S- β -CD.

DOX drug. The FESEM images revealed a cloud-like HS- β -CD coating with the gold nanorods embedded deep within it, shown in Fig. 8.

Dynamic light scattering (DLS) confirmed the successful functionalization of gold nanorods (AuNRs). The results are listed in Table 1. The attachment of β -cyclodextrin (β -CD-SH) increased the hydrodynamic size from 52.3 nm to 70.8 nm and reduced the surface charge from +39.7 mV to +29.9 mV. A further size increase of 40.6 nm and a drop in zeta potential to +24.1 mV upon doxorubicin (DOX) loading indicated the formation of AuNRs@S- β -CD-DOX nanocomposites. The significant size increase after DOX addition is likely due to hydrogen bonding between amine/hydroxyl groups of the drug and the cyclodextrin, which can induce aggregation.

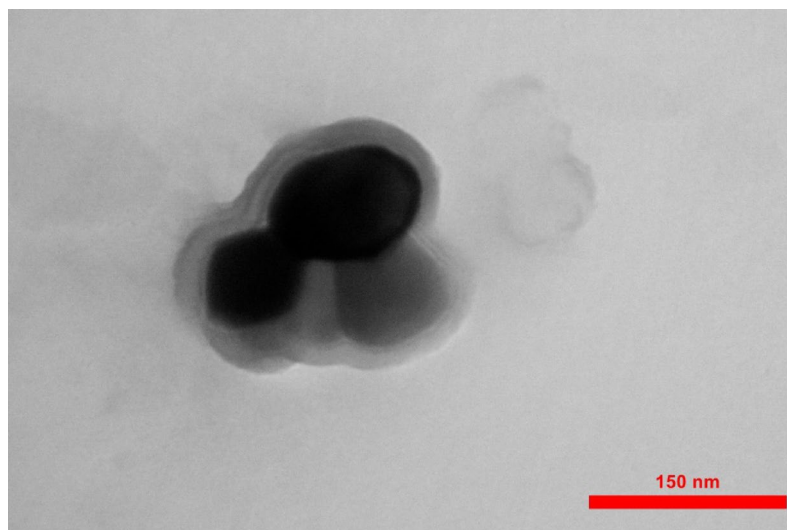


Fig. 7. TEM image of AuNR@S-β-CD-DOX.

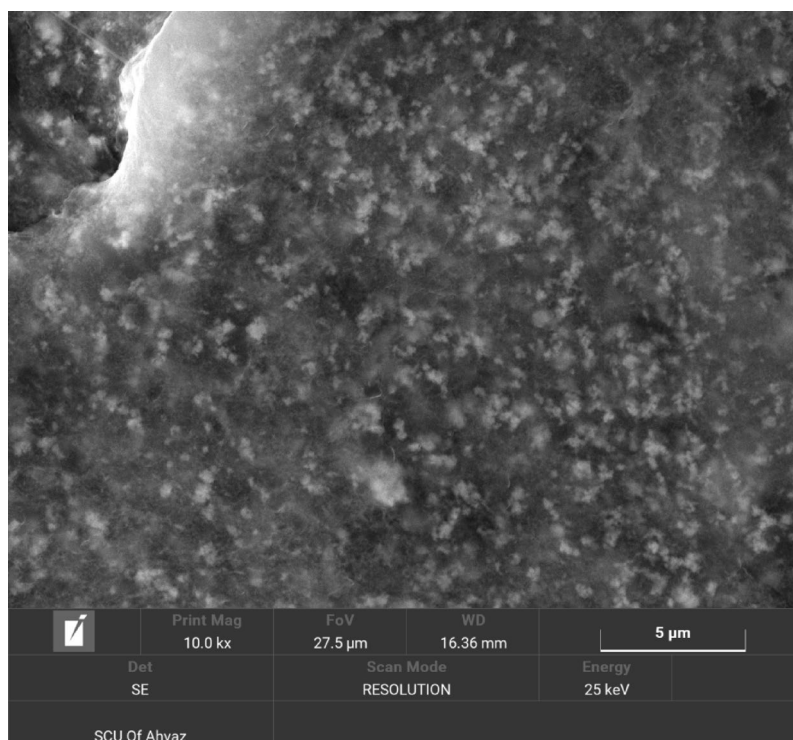


Fig. 8. FESEM image of AuNR@S-β-CD.

Sample	Hydrodynamic size (nm)	Zeta potential (mV)
AuNRs	52.3	+39.7
AuNRs@S-β-CD	70.8	+29.9
AuNRs@S-β-CD-DOX	111.4	+24.1

Table 1. Characterization of AuNRs, AuNRs@S-β-CD, and AuNRs@S-β-CD-DOX by dynamic light scattering method.

To load doxorubicin (DOX) onto the AuNRs@S- β -CD nanocomposite, a mixture of the drug and nanocomposite, in PBS buffer was stirred overnight. The resulting composite was then collected by centrifugation and purified. The loading efficiency was determined by measuring the unbound DOX in the supernatant using UV-Vis spectroscopy at 485 nm. The % drug loading efficiency (DLE %) of DOX was calculated by the following equation:

$$\text{DLE\%} = \frac{\text{Abs}_{(\text{original DOX})} - \text{Abs}_{(\text{residual DOX})}}{\text{Abs}_{(\text{original DOX})}} \times 100\%$$

The drug release profile was investigated under different pH conditions (5.5 and 7.4) with and without near-infrared (NIR) laser irradiation (3.6 W/cm² for 5 min). All samples were incubated at 37 °C with gentle shaking. At predetermined time intervals, the amount of DOX released was quantified by UV-Vis spectroscopy after centrifugation.

The porous nanostructure of the AuNRs@S- β -CD nanocomposite enabled efficient loading of doxorubicin (DOX) at 31.5 wt%. Drug release studies demonstrated a dual-responsive system, triggered by both pH and NIR laser irradiation. As the Fig. 9 shows, at a neutral pH of 7.4 (simulating blood circulation), DOX release was minimal due to its low solubility. In an acidic environment (pH 5.5, simulating cellular endosomes), release was enhanced as the protonation of β -CD hydroxyl groups disrupted electrostatic bonds with DOX, and solubility of the drug increased. The NIR laser irradiation significantly accelerated release. At pH 5.5, the cumulative DOX release reached 87% with irradiation versus 27% without after 24 h. Even at pH 7.4, irradiation boosted release to 48%, compared to 13% without irradiation. This enhancement is attributed to photothermal heating, which further dissociates the carrier-drug bonds. This dual-responsive mechanism ensures targeted drug delivery, enhancing anticancer efficacy by releasing DOX specifically in the acidic and laser-irradiated tumor microenvironment.

Fiber coupled diode laser

In this study, we employed fiber-coupled diode lasers with a nominal output power of 16 W each. To ensure experimental safety and prevent potential damage to the equipment, two identical lasers were used. Each laser was operated at an output power of approximately 8 W. The emission wavelength was 808 nm, and opto-mechanical components were utilized to direct the laser beams into the samples. The lasers were positioned above the samples so that their beams were aligned to converge precisely at a single focal point at the bottom of the sample container.

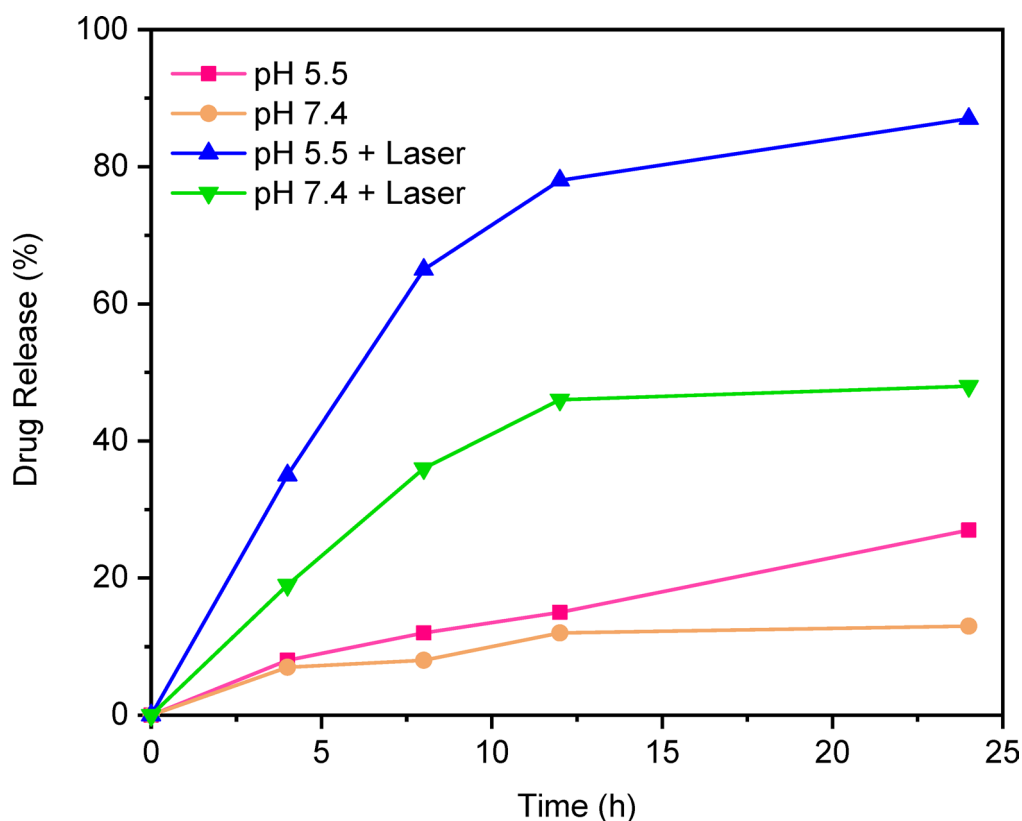


Fig. 9. Cumulative DOX release from AuNRs@S- β -CD-DOX with and without NIR laser irradiation at different pH values.

Gene	primer	Sequence	Annealing temp	Length
<i>Casp3</i>	Reverse	TCTTTAGAAACATCACGCATC	48.5	147 bp
	Forward	AAGCACTGGAATGACATCTC	49.7	
<i>Casp8</i>	Reverse	CTTTGCTGAATTCTTCATAGTCG	51.7	115 bp
	Forward	ATGGAGAAGAGGGTCATCC	51.1	
<i>Casp9</i>	Reverse	GGTCTCAACGTACCAGGAGC	55.9	177 bp
	Forward	CCTGGCAGTAACCCCGAG	54.9	
<i>B-act</i>	Reverse	GGGACTTCCTGTAACAACGCA	61	103 bp
	Forward	GAGCATCCCCAAAGTTCACA	61	

Table 2. The primers used as well as their sequences and product lengths are presented here.

Prior to the final experiments, the temperature of the samples was measured using a digital thermocouple thermometer. Care was taken to prevent the laser beam from directly hitting the sensor head, which could result in artificially elevated temperature readings. A total laser power of 16 W was delivered to each sample, as confirmed by a power meter, and we ensured that this same power level was consistently applied in all irradiation procedures. The lasers were actively cooled using water circulation and a chiller system, maintaining the laser body at approximately 4 °C.

Cell culture

The A549 cell line was obtained from the cell bank of the Pasteur Institute of Iran. For all experimental conditions discussed, the cells were cultured in a 5% carbon dioxide incubator at 95% humidity, using DMEM medium supplemented with 10% FBS and 1% penicillin-streptomycin. A total of 50,000 A549 cells were seeded in each well of a 12-well plate and divided into 8 treatment groups. The treatment groups were as follows: control (cont), gold nanorods modified with β -cyclodextrin (AuNRs@S- β -CD), NIR laser 808 nm (laser), doxorubicin (DOX(1.307)), doxorubicin-loaded gold nanorods (DOX(0.078)+AuNRs@S- β -CD), PTT with gold nanorods+NIR laser 808 nm (AuNRs@S- β -CD+laser), doxorubicin+NIR laser 808 nm (DOX(1.307)+laser), and chemo-photothermal treatment with doxorubicin-loaded gold nanorods+NIR laser 808 nm (DOX(0.078)+AuNRs@S- β -CD+laser).

In the referenced study, utilizing data from previous research by Deinavizadeh et al., the cell viability changes under AuNRs@S- β -CD and DOX(0.078)+AuNRs@S- β -CD conditions, before and after laser exposure, showed optimal results at a concentration of 3/125 nm. Consequently, this concentration, which is approximately half of the IC₅₀ for these materials, was selected for gene expression studies and other experiments. Treated cells from the AuNRs@S- β -CD, DOX(1.307), and DOX(0.078)+AuNRs@S- β -CD groups were evaluated after 48 h of treatment, while the laser, AuNRs@S- β -CD+laser, DOX(1.307)+laser, and DOX(0.078)+AuNRs@S- β -CD+laser groups were examined 20 min post-laser exposure.

Gene expression

Cells were seeded in 12-well plates according to the mentioned groupings and treated accordingly. The cells were then collected from the plates using Trizol and total RNA was extracted following the manufacturer's protocols. The quality and quantity of the extracted RNA were assessed using a NanoDrop device based on the 260/280 and 260/230 ratios and were also analyzed on a 1% agarose gel. Subsequently, cDNA was synthesized using the cDNA synthesis kit from Yekta Tajhiz. The primers utilized are listed in the Table 2. Quantitative PCR was performed using the SYBR Green kit from Yekta Tajhiz and the ABI 7500 real-time PCR system (ABI, USA), with the resulting data analyzed using the $2^{-\Delta CT}$ formula. β -actin amplification was used as the housekeeping gene.

Western blotting

The seeded cells were treated according to their respective groups and harvested after 48 h. They were then lysed using a lysis buffer (Tris-HCl, EDTA, NaCl, Sodium Deoxycholate, SDS, Protease Inhibitor Cocktail, and NP-40 (1% Triton)) to isolate the proteins. Proteins were separated on an SDS-PAGE polyacrylamide gel and subsequently transferred to PVDF membranes using a Western blot apparatus and transfer buffer at 120 mV for one and a half hours. To prevent non-specific reactions, a blocking solution was applied, and the proteins on the PVDF membrane were incubated for one hour with the blocking solution. Following the blocking step, the PVDF membrane was incubated for 16–18 h at room temperature with the primary antibody. The membrane was then washed three times with TBS-T buffer and incubated for one hour with the secondary antibody at room temperature. Detection was performed using ECL advanced reagents.

TUNEL assay

Cells were processed as described previously and treated according to the mentioned groupings after 24 h. The culture medium was removed, and the cells were incubated and fixed with 4% formaldehyde for 20 min at 4 °C. The samples were then washed with PBS. The samples were treated with X1 TBS solution (Sigma 5912-T) at boiling point for 20 min. After washing with PBS, Triton 3% was added for 30 min. To stop the reaction, the samples were exposed to secondary goat serum antibody (10%) for 45 min. After washing, the samples were incubated with the primary antibody for 24 h at 2–8 °C. Following additional washing, the samples were incubated with the secondary antibody for 90 min at 37 °C.

Ethics approval and consent to participate

All authors agree to the ethics and consent to participate in this article and declare that this submission follows the policies of *Scientific Reports*. Accordingly, the material is the authors' original work, which has not been previously published elsewhere. The paper is not being considered for publication elsewhere. All authors have been personally and actively involved in substantial work leading to the paper and will take public responsibility for its content.

Data availability

The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request. No specific repository deposition was required as the data fall outside the mandatory categories defined by the journal.

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Declarations

Competing interests

The authors declare no competing interests.

Additional information

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